5 Computer Networks (132)

5.0.1 ISRO CSE 2014 | Question: 58

Assume the following information.

- Original timestamp value $= 46$
- Receive timestamp value $= 59$
- Transmit timestamp value $= 60$
- Timestamp at arrival of packet $= 69$

Which of the following statements is correct?

A. Receive clock should go back by 3 milliseconds
B. Transmit and Receive clocks are synchronized
C. Transmit clock should go back by 3 milliseconds
D. Receive clock should go ahead by 1 milliseconds

isro2014 computer-networks icmp

Answer key

5.0.2 ISRO-2013-19

What is the maximum number of characters (7 bits + parity) that can be transmitted in a second on a 19.2 Kbps line. This asynchronous transmission requires 1 start bit and 1 stop bit.

A. 192 B. 240 C. 1920 D. 1966

isro2013 serial-communication

Answer key

5.0.3 ISRO-2013-44

Which algorithm is used to shape the bursty traffic into a fixed rate traffic by averaging the data rate?

A. Solid bucket algorithm
B. Spanning tree algorithm
C. Hocken helm algorithm
D. Leaky bucket algorithm

isro2013 computer-networks

Answer key

5.0.4 ISRO CSE 2009 | Question: 5

What is the primary purpose of a VLAN?

A. Demonstrating the proper layout for a network
B. Simulating a network
C. To create a virtual private network
D. Segmenting a network inside a switch or device

isro2009 computer-networks

Answer key

5.0.5 ISRO CSE 2007 | Question: 19

Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through a synchronous serial line at 2400 baud rate, and with two stop bits is

A. 109 B. 216 C. 218 D. 219

isro2007 serial-communication

Answer key☟

5.0.6 ISRO2015-51

How many characters per sec $(7 \text{ bits} + 1 \text{ parity})$ can be transmitted over a 2400 bps line if the transfer is synchronous $(1$ start and 1 stop bit)?

A. 300 B. 240 C. 250 D. 275

isro2015 computer-networks serial-communication

Answer key☟

5.0.7 ISRO-DEC2017-33

Assume that Source S and Destination D are connected through an intermediate router R.
How many times a packet has to visit the network layer and data link layer during a transmission from S to D ?

A. Network layer-4 times, Data link layer-4 times
B. Network layer-4 times, Data link layer-6 times
C. Network layer-2 times, Data link layer-4 times
D. Network layer-3 times, Data link layer-4 times

isrodec2017 computer-networks

Answer key☟

5.0.8 ISRO CSE 2023 | Question: 80

Match the following and select the correct option:

(i)	Encryption	(P)	Manipulation of identification of a host (through NIC identifier)
(ii)	MAC cloning	(Q)	Determining version and name of OS and OS utilities
(iii)	ARP spoofing	(R)	Protecting the confidentiality
(i)	Finger printing	(S)	Poisioning the IP to MAC address mapping

A. (i) – (R), (iI) – (P), (i) – (S), (i) – (Q)
B. (i) – (P), (iI) – (R), (i) – (S), (i) – (Q)
C. (i) – (R), (iI) – (S), (i) – (P), (i) – (Q)
D. (i) – (R), (iI) – (Q), (i) – (S), (i) – (P)

isro-cse-2023 computer-networks match-the-following

5.0.9 ISRO CSE 2023 | Question: 65

For internal data communication through computer network, a 5 layer protocol hierarchy is proposed. Applications to be used in this network generate 270 bytes long messages. At each of the layers, a 6 -byte header is added. What fraction of network bandwidth(in \%) is wasted due to headers?

A. 5% B. 20% C. 10% D. 15%

isro-cse-2023 computer-networks

5.0.10 ISRO CSE 2023 | Question: 59

Which of the following is the cause of Ping of death issue related to ICMP packets?

A. Buffer overflow
B. Divide by ZERO
C. Missing input sanitisation
D. Privilege escalation

isro-cse-2023 computer-networks icmp

Answer key☟

5.0.11 ISRO CSE 2023 | Question: 17

Match the following context of information Secuirty which are closely associated

Matching

(i)	Ingress Filtering	(P)	Data Leakage Prevention
(ii)	Egress Filtering	(Q)	Hiding indentity of systems
(iii)	NAT	(R)	Keep Track of TCP/IP connections
(iv)	Stateful firewall	(S)	Malicious traffic prevention

A. (i) - (S), (ii) - (P), (iii) - (Q), (iv) - (R)
B. (i) - (P), (ii) - (R), (iii) - (S), (iv) - (Q)
C. (i) - (S), (ii) - (R), (iii) - (P), (iv) - (Q)
D. (i) - (R), (ii) - (Q), (iii) - (S), (iv) - (P)

isro-cse-2023 computer-networks match-the-following

Answer key☟

5.1 Application Layer Protocols (1)

5.1.1 Application Layer Protocols: GATE CSE 2008 | Question: 14, ISRO2016-74

What is the maximum size of data that the application layer can pass on to the TCP layer below?

A. Any size
B. 2^{16} bytes - size of TCP header
C. 2^{16} bytes
D. 1500 bytes

gatecse-2008 easy computer-networks application-layer-protocols isro2016

Answer key☟

5.2 CRC Polynomial (4)

5.2.1 CRC Polynomial: GATE CSE 2007 | Question: 68, ISRO2016-73

The message 11001001 is to be transmitted using the CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is:

A. 11001001000
B. 11001001011
C. 11001010
D. 110010010011

gatecse-2007 computer-networks error-detection crc-polynomial normal isro2016

Answer key☟

5.2.2 CRC Polynomial: ISRO CSE 2018 | Question: 11

________ can detect burst error of length less than or equal to degree of the polynomial and detects burst errors that affect odd number of bits.

A. Hamming Code
B. CRC
C. VRC
D. None of the above

isro2018 computer-networks crc-polynomial

Answer key☟

5.2.3 CRC Polynomial: ISRO-2013-40

If the frame to be transmitted is 1101011011 and the CRC polynomial to be used for generating checksum is $x^4 + x + 1$, than what is the transmitted frame?

A. 11010110111011
B. 11010110111101
C. 11010110111110
D. 11010110111001

isro2013 computer-networks crc-polynomial

Answer key☟

5.2.4 CRC Polynomial: ISRO2015-52

In CRC if the data unit is 100111001 and the divisor is 1011 then what is dividend at the receiver?

A. 100111001101
B. 100111001011
C. 100111001
D. 100111001110

isro2015 computer-networks crc-polynomial

Answer key☟

5.3 CSMA CD (2)

5.3.1 CSMA CD: ISRO CSE 2008 | Question: 6

Which of the following transmission media is not readily suitable to CSMA operation?

A. Radio
B. Optical fibers
C. Coaxial cable
D. Twisted pair

isro2008 computer-networks csma-cd

Answer key☟

5.3.2 CSMA CD: ISRO CSE 2016 | Question: 69

In Ethernet CSMA/CD, the special bit sequence transmitted by media access management to handle collision is called

A. Preamble
B. Postamble
C. Jam
D. None of these

computer-networks csma-cd isro2016

Answer key☟

5.4 Communication (7)

5.4.1 Communication: ISRO CSE 2007 | Question: 20

If the bandwidth of a signal is 5 kHz and the lowest frequency is 52 kHz, what is the highest frequency

A. 5 kHz B. 10 kHz C. 47 kHz D. 57 kHz

isro2007 communication

Answer key

5.4.2 Communication: ISRO CSE 2007 | Question: 22

Phase transition for each bit are used in

A. Amplitude modulation
B. Carrier modulation
C. Manchester encoding
D. NRZ encoding

isro2007 computer-networks communication

Answer key

5.4.3 Communication: ISRO CSE 2007 | Question: 24, ISRO CSE 2016 | Question: 67

Bit stuffing refers to

A. inserting a 0 in user stream to differentiate it with a flag
B. inserting a 0 in flag stream to avoid ambiguity
C. appending a nipple to the flag sequence
D. appending a nipple to the use data stream

isro2007 communication isro2016

Answer key

5.4.4 Communication: ISRO CSE 2008 | Question: 9

What is the bandwidth of the signal that ranges from 40 kHz 4 MHz

A. 36 MHz B. 360 kHz C. 3.96 MHz D. 396 kHz

isro2008 communication non-gatecse

Answer key

5.4.5 Communication: ISRO CSE 2011 | Question: 1

The encoding technique used to transmit the signal in giga ethernet technology over fiber optic medium is

A. Differential Manchester encoding
B. Non return to zero
C. $4B/5B$ encoding
D. $8B/10B$ encoding

computer-networks isro2011 communication

Answer key

5.4.6 Communication: ISRO CSE 2014 | Question: 45

Consider a 50 kbps satellite channel with a 500 milliseconds round trip propagation delay. If the sender wants to transmit 1000 bit frames, how much time will it take for the receiver to receive the frame?

A. 250 milliseconds
B. 20 milliseconds
C. 520 milliseconds
D. 270 milliseconds

isro2014 computer-networks communication

Answer key

5.4.7 Communication: ISRO2015-49

What frequency range is used for microwave communications, satellite and radar?

A. Low frequency: 30 kHz to 300 kHz
B. Medium frequency: 300 kHz to 3 MHz
C. Super high frequency: 3000 MHz to 30000 MHz
D. Extremely high frequency: 30000 kHz

isro2015 computer-networks communication

Answer key

5.5 Cryptography (8)

5.5.1 Cryptography: ISRO CSE 2009 | Question: 7

Advanced Encryption Standard (AES) is based on

A. Asymmetric key algorithm
B. Symmetric key algorithm
C. Public key algorithm
D. Key exchange

isro2009 computer-networks cryptography network-security

Answer key

5.5.2 Cryptography: ISRO CSE 2017 | Question: 35

MD5 is a widely used hash function for producing hash value of

A. 64 bits
B. 128 bits
C. 512 bits
D. 1024 bits

isro2017 computer-networks network-security cryptography

Answer key

5.5.3 Cryptography: ISRO CSE 2018 | Question: 78

Which one of the following algorithm is not used in asymmetric key cryptography?

a. RSA Algorithm
b. Gillie-Hellman Algorithm
c. Electronic Code Book Algorithm
d. None of the above

isro2018 computer-networks network-security cryptography

Answer key

5.5.4 Cryptography: ISRO CSE 2020 | Question: 44

In a columnar transportation cipher, the plain text is “the tomato is a plant in the night shade family”, keyword is “TOMATO”. The cipher text is

A. $\text{"TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"}$
B. $\text{"TINESAX / EOAHTFX / MAIIAIX / HTLTHEY / TAPNGDL / OSTNHMX"}$
C. $\text{"TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / OSTNHMX / TAPNGDL"}$
D. $\text{"EOAHTFX / TINESAX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"}$

isro-2020 computer-networks cryptography normal

Answer key

5.5.5 Cryptography: ISRO CSE 2020 | Question: 45

Avalanche effect in cryptography refers

A. Large changes in cipher text when the keyword is changed minimally
B. Large changes in cipher text when the plain text is changed
C. Large Impact of keyword change to length of the cipher text
D. None of the above

isro-2020 computer-networks cryptography normal

Answer key☟

5.5.6 Cryptography: ISRO-2013-21

What will be the cipher text produced by the following cipher function for the plain text ISRO with key $k = 7$. [Consider $'A' = 0,' B' = 1, \dots ' Z' = 25$]

$C_k(M) = (kM + 13) \mod 26$

A. RJCH B. QIBG C. GQPM D. XPIN

isro2013 cryptography

Answer key☟

5.5.7 Cryptography: ISRO-2013-46

Which of the following encryption algorithms is based on the Feistal structure?

A. Advanced Encryption Standard
B. RSA public key cryptographic algorithm
C. Data Encryption standard
D. RC4

isro2013 cryptography

Answer key☟

5.5.8 Cryptography: ISRO-DEC2017-29

Using public key cryptography, X adds a digital signature σ to a message M, encrypts $\langle M, \sigma \rangle$ and sends it to Y, where it is decrypted. Which one of the following sequence of keys is used for operations ?

A. Encryption: $X's$ private key followed by $Y's$ private key; Decryption:$X's$ public key followed by $Y's$ public key.
B. Encryption: $X's$ private key followed by $Y's$ public key; Decryption:$X's$ public key followed by $Y's$ private key.
C. Encryption: $X's$ private key followed by $Y's$ public key; Decryption:$Y's$ private key followed by $X's$ public key.
D. Encryption: $X's$ public key followed by $Y's$ private key; Decryption:$Y's$ public key followed by $X's$ private key.

isrodec2017 network-security cryptography

Answer key☟

5.6 Data Transfer Speed (1)

5.6.1 Data Transfer Speed: ISRO CSE 2023 | Question: 45

In a computer network using CSMA/CD, the minimum frame size for data transfer is 1500 bytes. What will be the data transfer speed in this network over 1 Km 375 cable, if the signal speed in the cable is $250000 \mathrm{Km}/\mathrm{sec}$ and no repeaters are used in this network?

A. 1 Gbps B. 100 Mbps C. 500 Mbps D. 1.5 Gbps

isro-cse-2023 csma-cd computer-networks data-transfer-speed

Answer key☟

5.7 Distance Vector Routing (1)

5.7.1 Distance Vector Routing: ISRO CSE 2017 | Question: 34

Match with the suitable one :

	List-1		List-2
(A)	Multicast group membership	1.	Distance Vector routing
(B)	Interior gateway protocol	2.	IGMP
(C)	Exterior gateway protocol	3.	OSPF
(D)	RIP	4.	BGP

A. A-2, B-3, C-4, D-1
B. A-2, B-4, C-3, D-1
C. A-3, B-4, C-1, D-2
D. A-3, B-1, C-4, D-2

isro2017 computer-networks distance-vector-routing

Answer key☟

5.8 Dns (2)

5.8.1 Dns: ISRO CSE 2016 | Question: 80

When a DNS server accepts and uses incorrect information from a host that has no authority giving that information, then it is called

A. DNS lookup
B. DNS hijacking
C. DNS spoofing
D. None of the mentioned

isro2016 computer-networks dns

Answer key☟

5.8.2 Dns: ISRO2015-44

The DNS maps the IP address to

A. A binary address as strings
B. A n alphanumeric address
C. A hierarchy of domain names
D. A hexadecimal address

isro2015 computer-networks dns

Answer key☟

5.9 Encoding (2)

5.9.1 Encoding: ISRO CSE 2007 | Question: 13

By using an eight bit optical encoder the degree of resolution that can be obtained is (approximately)

A. 1.8° B. 3.4° C. 2.8° D. 1.4°

isro2007 communication encoding

Answer key☟

5.9.2 Encoding: ISRO CSE 2020 | Question: 52

To send same bit sequence, NRZ encoding require

A. Same clock frequency as Manchester encoding
B. Half the clock frequency as Manchester encoding
C. Twice the clock frequency as Manchester encoding
D. A clock frequency which depend on number of zeroes and ones in the bit sequence

isro-2020 computer-networks encoding normal

Answer key☟

5.10 Error Correction (2)

5.10.1 Error Correction: ISRO CSE 2011 | Question: 48

The hamming distance between the octets of 0xAA and 0x55 is

A. 7 B. 5 C. 8 D. 6

isro2011 computer-networks error-correction hamming-code

Answer key☟

5.10.2 Error Correction: ISRO CSE 2023 | Question: 3

Which of the following is true?

A. Both Parity and Cyclic Redundancy Check are error correcting codes
B. Both Parity and Low Density Parity Check Code are error correcting codes
C. Both Reed-Solomon code and Low Density Parity Check Code are error correcting codes
D. Both Cyclic Redundancy Check and Low Density Parity Check Code are error correcting codes

isro-cse-2023 error-correction computer-networks

Answer key☟

5.11 Error Detection (4)

5.11.1 Error Detection: ISRO CSE 2008 | Question: 79

Repeated execution of simple computation may cause compounding of

A. round-off errors
B. syntax errors
C. run-time errors
D. logic errors

isro2008 computer-networks error-detection

Answer key☟

5.11.2 Error Detection: ISRO CSE 2011 | Question: 50

Data is transmitted continuously at 2.048 Mbps rate for 10 hours and received 512 bits errors. What is the bit error rate?

A. 6.9 e-9 B. 6.9 e-6 C. 69 e-9 D. 4 e-9

isro2011 computer-networks error-detection

Answer key☟

5.11.3 Error Detection: ISRO CSE 2023 | Question: 26

Which of the following methods is used to detect double errors?

A. Odd Parity
B. Even Parity
C. Checksum (CRC)
D. Checksum (XOR)

isro-cse-2023 computer-networks error-detection

Answer key☟

5.11.4 Error Detection: ISRO-2013-18

How many check bits are required for 16 bit data word to detect 2 bit errors and single bit correction using hamming code?

A. 5 B. 6 C. 7 D. 8

isro2013 error-detection

Answer key☟

5.12 Ethernet (6)

5.12.1 Ethernet: ISRO CSE 2007 | Question: 21

An Ethernet hub

A. functions as a repeater
B. connects to a digital PBX
C. connects to a token-ring network
D. functions as a gateway

isro2007 computer-networks ethernet

Answer key☟

5.12.2 Ethernet: ISRO CSE 2008 | Question: 5

In Ethernet, the source address field in the MAC frame is the _______ address.

A. original sender's physical
B. previous station's physical
C. next destination's physical
D. original sender's service port

isro2008 computer-networks ethernet

Answer key☟

5.12.3 Ethernet: ISRO CSE 2014 | Question: 61

A mechanism or technology used in Ethernet by which two connected devices choose common transmission parameters such as speed, duplex mode and flow control is called

A. Autosense B. Synchronization C. Pinging D. Auto negotiation

isro2014 computer-networks ethernet

Answer key☟

5.12.4 Ethernet: ISRO CSE 2017 | Question: 33

An Ethernet frame that is less than the IEEE 802.3 minimum length of 64 octets is called

A. Short frame
B. Small frame
C. Mini frame
D. Runt frame

isro2017 computer-networks ethernet

Answer key☟

5.12.5 Ethernet: ISRO-2013-38

In the Ethernet, which field is actually added at the physical layer and is not part of the frame.

A. Preamble
B. CRC
C. Address
D. Location

isro2013 computer-networks ethernet

Answer key☟

5.12.6 Ethernet: ISRO-2013-39

Ethernet layer- 2 switch is a network element type which gives.

A. Different collision domain and same broadcast domain.
B. Different collision domain and different broadcast domain.
C. Same collision domain and same broadcast domain.
D. Same collision domain and different broadcast domain.

isro2013 computer-networks ethernet

Answer key☟

5.13 Firewall (1)

5.13.1 Firewall: ISRO CSE 2018 | Question: 77

What is one advantage of setting up a DMZ (Demilitarized Zone) with two firewalls?

A. You can control where traffic goes in the three networks
B. You can do statefull packet filtering
C. You can do load balancing
D. Improve network performance

isro2018 computer-networks network-security firewall

Answer key☟

5.14 IP Addressing (3)

5.14.1 IP Addressing: ISRO CSE 2008 | Question: 32

The network $198.78.41.0$ is a

A. Class A network
B. Class B network
C. Class C network
D. Class D network

isro2008 computer-networks ip-addressing

Answer key☟

5.14.2 IP Addressing: ISRO CSE 2014 | Question: 6

The process of modifying IP address information in IP packet headers while in transit across a traffic routing device is called

A. Port address translation (PAT)
B. Network address translation (NAT)

C. Address mapping

D. Port mapping

isro2014 computer-networks ip-addressing

Answer key

5.14.3 IP Addressing: ISRO-DEC2017-31

In the IPv4 addressing format, the number of networks allowed under Class C addresses is

A. 2^{20}
B. 2^{24}
C. 2^{14}
D. 2^{21}

isrodec2017 computer-networks ip-addressing

Answer key

5.15 IP Packet (3)

5.15.1 IP Packet: ISRO CSE 2009 | Question: 74

Use of IPSEC in tunnel mode results in

A. IP packet with same header
B. IP packet with new header
C. IP packet without header
D. No changes in IP packet

isro2009 computer-networks ip-packet

Answer key

5.15.2 IP Packet: ISRO CSE 2014 | Question: 31

A IP packet has arrived in which the fragmentation offset value is 100, the value of HLEN is 5 and the value of total length field is 200. What is the number of the last byte?

A. 194
B. 394
C. 979
D. 1179

computer-networks ip-packet isro2014

Answer key

5.15.3 IP Packet: ISRO CSE 2014 | Question: 55

An IP packet has arrived with the first 8 bits as 01000010. Which of the following is correct?

A. The number of hops this packet can travel is 2
B. The total number of bytes in header is 16 bytes
C. The upper layer protocol is ICMP
D. The receiver rejects the packet

isro2014 computer-networks ip-packet

Answer key

5.16 Inter Process Communication (1)

5.16.1 Inter Process Communication: ISRO CSE 2020 | Question: 58

Remote Procedure Calls are used for

A. communication between two processes remotely different from each other on the same system
B. communication between two processes on the same system
C. communication between two processes on the separate systems
D. none of the above

isro-2020 computer-networks inter-process-communication normal

Answer key☟

5.17 Ipv6 (1)

5.17.1 Ipv6: ISRO-2013-42

IPv6 does not support which of the following addressing modes?

A. Unicast addressing
B. Multicast addressing
C. Broadcast addressing
D. Anycast addressing

isro2013 computer-networks ipv6

Answer key☟

5.18 LAN Technologies (1)

5.18.1 LAN Technologies: ISRO CSE 2008 | Question: 4

On a LAN ,where are IP datagrams transported?

A. In the LAN header
B. In the application field
C. In the information field of the LAN frame
D. After the TCP header

isro2008 computer-networks lan-technologies

Answer key☟

5.19 Link State Routing (1)

5.19.1 Link State Routing: ISRO CSE 2007 | Question: 26

If there are five routers and six networks in intranet using link state routing, how many routing tables are there?

A. 1
B. 5
C. 6
D. 11

isro2007 computer-networks routing link-state-routing

Answer key☟

5.20 MAC Protocol (4)

5.20.1 MAC Protocol: ISRO CSE 2009 | Question: 4

Which of the following is a MAC address?

A. 192.166.200.50
B. 00056A:01A01A5CCA7FF60
C. 568, Airport Road
D. 01:A5:BB:A7:FF:60

isro2009 computer-networks mac-protocol

Answer key☟

5.20.2 MAC Protocol: ISRO CSE 2011 | Question: 20

The IEEE standard for WiMax technology is

A. IEEE 802.16
B. IEEE 802.36
C. IEEE 812.16
D. IEEE 806.16

isro2011 non-gatecse computer-networks mac-protocol

Answer key☟

5.20.3 MAC Protocol: ISRO CSE 2014 | Question: 71

Which of the following is not a valid multicast MAC address?

A. 01:00:5E:00:00:00
B. 01:00:5E:00:00:FF
C. 01:00:5E:00:FF:FF
D. 01:00:5E:FF:FF:FF

computer-networks mac-protocol isro2014

Answer key

5.20.4 MAC Protocol: ISRO CSE 2017 | Question: 32

Which media access control protocol is used by IEEE 802.11 wireless LAN?

A. CDMA
B. CSMA/CA
C. ALOHA
D. None of the above

isro2017 computer-networks mac-protocol

Answer key

5.21 Network Layer (2)

5.21.1 Network Layer: ISRO CSE 2009 | Question: 2

In networking, UTP stands for

A. Unshielded T-connector port
B. Unshielded twisted pair
C. Unshielded terminating pair
D. Unshielded transmission process

isro2009 computer-networks network-layer

Answer key

5.21.2 Network Layer: ISRO CSE 2017 | Question: 28

In networking terminology UTP means

A. Uniquitous teflon port
B. Uniformly terminating port
C. Unshielded twisted pair
D. Unshielded T-connector port

isro2017 computer-networks network-layer

Answer key

5.22 Network Layering (3)

5.22.1 Network Layering: ISRO CSE 2007 | Question: 75

When a host on network A sends a message to a host on network B, which address does the router look at?

A. Port
B. IP
C. Physical
D. Subnet mask

isro2007 computer-networks network-layering

Answer key

5.22.2 Network Layering: ISRO CSE 2011 | Question: 32

In which layer of network architecture, the secured socket layer (SSL) is used?

A. physical layer
B. session layer
C. application layer
D. presentation layer

isro2011 computer-networks network-security network-layering

Answer key

5.22.3 Network Layering: ISRO2015-58

Which layers of the OSI reference model are host-to-host layers?

A. Transport, session, presentation, application
B. Session, presentation, application
C. Datalink, transport, presentation, application
D. Physical, datalink, network, transport

isro2015 computer-networks network-layering

Answer key☟

5.23 Network Protocols (11)

5.23.1 Network Protocols: ISRO CSE 2007 | Question: 63

Which of these is not a feature of WAP 2.0

A. Push and Pull Model
B. Interface to a storage device
C. Multimedia messaging
D. Hashing

isro2007 network-protocols

Answer key☟

5.23.2 Network Protocols: ISRO CSE 2008 | Question: 10

Which Project 802 standard provides for a collision-free protocol?

A. 802.2
B. 802.3
C. 802.5
D. 802.6

isro2008 computer-networks network-protocols

Answer key☟

5.23.3 Network Protocols: ISRO CSE 2009 | Question: 3

The address resolution protocol (ARP) is used for

A. Finding the IP address from the DNS
B. Finding the IP address of the default gateway
C. Finding the IP address that corresponds to a MAC address
D. Finding the MAC address that corresponds to an IP address

isro2009 computer-networks network-protocols

Answer key☟

5.23.4 Network Protocols: ISRO CSE 2011 | Question: 12

The network protocol which is used to get MAC address of a node by providing IP address is

A. SMTP
B. ARP
C. RIP
D. BOOTP

isro2011 computer-networks network-protocols

Answer key☟

5.23.5 Network Protocols: ISRO CSE 2011 | Question: 69

Lightweight Directory Access protocol is used for

A. Routing the packets
B. Authentication
C. obtaining IP address
D. domain name resolving

isro2011 computer-networks network-protocols

Answer key

5.23.6 Network Protocols: ISRO CSE 2016 | Question: 70

Which network protocol allows hosts to dynamically get a unique IP number on each bootup

A. DHCP
B. BOOTP
C. RARP
D. ARP

isro2016 computer-networks network-protocols

Answer key

5.23.7 Network Protocols: ISRO CSE 2017 | Question: 31

Which of the following protocol is used for transferring electronic mail messages from one machine to another?

A. TELNET
B. FTP
C. SNMP
D. SMTP

isro2017 computer-networks network-protocols

Answer key

5.23.8 Network Protocols: ISRO-2013-47

The protocol data unit for the transport layer in the internet stack is

A. Segment
B. Message
C. Datagram
D. Frame

isro2013 network-protocols

Answer key

5.23.9 Network Protocols: ISRO-DEC2017-30

Which of the following are used to generate a message digest by the network security protocols?

(P) SHA-256
(Q) AES
(R) DES
(S) MD5

A. P and S only
B. P and Q only
C. R and S only
D. P and R only

isrodec2017 network-security network-protocols

Answer key

5.23.10 Network Protocols: ISRO-DEC2017-35

Consider the set of activities related to e-mail

A: Send an e-mail from a mail client to mail server.
B: Download e-mail headers from mailbox and retrieve mails from server to cache
C: Checking e-mail through a web browser

The application level protocol used for each activity in the same sequence is

A. SMTP,HTTPS,IMAP

B. SMTP,POP,IMAP

C. SMTP,IMAP,HTTPS

D. SMTP,IMAP,POP

isrodec2017 computer-networks application-layer-protocols network-protocols

Answer key☟

5.23.11 Network Protocols: ISRO2015-56

Which statement is false?

A. PING is a TCP/IP application that sends datagrams once every second in the hope of an echo response from the machine being PINGED
B. If the machine is connected and running a TCP/IP protocol stack, it should respond to the PING datagram with a datagram of its own
C. If PING encounters an error condition, an ICMP message is not returned
D. PING display the time of the return response in milliseconds or one of several error message

isro2015 computer-networks network-protocols

Answer key☟

5.24 Network Security (12)

5.24.1 Network Security: ISRO CSE 2007 | Question: 68

SSL is not responsible for

A. Mutual authentication of client & server
B. Secret communication
C. Data Integrity protection
D. Error detection and correction

isro2007 computer-networks network-security

Answer key☟

5.24.2 Network Security: ISRO CSE 2007 | Question: 70

The standard for certificates used on internet is

A. X.25
B. X.301
C. X.409
D. X.509

isro2007 computer-networks network-security

Answer key☟

5.24.3 Network Security: ISRO CSE 2007 | Question: 71

Hashed message is signed by a sender using

A. his public key
B. his private key
C. receiver's public key
D. receiver's private key

isro2007 computer-networks network-security

Answer key☟

5.24.4 Network Security: ISRO CSE 2008 | Question: 59

A public key encryption system

A. allows anyone to decode the transmissions
B. allows only the correct sender to decode the data

C. allows only the correct receiver to decode the data
D. does not encode the data before transmitting it

isro2008 computer-networks network-security

Answer key

5.24.5 Network Security: ISRO CSE 2009 | Question: 6

SHA-1 is a

A. encryption algorithm
B. decryption algorithm
C. key exchange algorithm
D. message digest function

isro2009 computer-networks network-security

Answer key

5.24.6 Network Security: ISRO CSE 2011 | Question: 18

An example of poly-alphabetic substitution is

A. P-box
B. S-box
C. Caesar cipher
D. Vigenere cipher

isro2011 computer-networks network-security

Answer key

5.24.7 Network Security: ISRO CSE 2014 | Question: 24

In a system an RSA algorithm with $p = 5$ and $q = 11$, is implemented for data security. What is the value of the decryption key if the value of the encryption key is 27?

A. 3
B. 7
C. 27
D. 40

isro2014 computer-networks network-security

Answer key

5.24.8 Network Security: ISRO CSE 2017 | Question: 36

Which protocol suite designed by IETF to provide security for a packet at the Internet layer?

A. IPSec
B. NetSec
C. PacketSec
D. SSL

isro2017 computer-networks network-security

Answer key

5.24.9 Network Security: ISRO CSE 2017 | Question: 37

Pretty Good Privacy (PGP) is used in:

A. Browser security
B. FTP security
C. Email security
D. None of the above

isro2017 computer-networks network-security

Answer key

5.24.10 Network Security: ISRO CSE 2018 | Question: 46

In cryptography, the following uses transposition ciphers and the keyword is LAYER. Encrypt the following message. (Spaces are omitted during encrypton)

$\text{WELCOME TO NETWORK SECURITY!}$

a. WMEKREETSILTWETCOOCYONRU!
b. EETSICOOCYWMEKRONRU!LTWET
c. LTWETONRU!WMEKRCOOCYEETSI
d. ONRU!COOCYLTWETEETSIWMEKR

isro2018 computer-networks network-security

Answer key☟

5.24.11 Network Security: ISRO CSE 2018 | Question: 74

Avalanche effect in cryptography

a. Is desirable property of cryptographic algorithm
b. Is undesirable property of cryptographic algorithm
c. Has no effect on encryption algorithm
d. None of the above

isro2018 computer-networks network-security

Answer key☟

5.24.12 Network Security: ISRO-2013-45

A packet filtering firewall can

A. Deny certain users from accessing a service
B. Block worms and viruses from entering the network
C. Disallow some files from being accessed through FTP
D. Block some hosts from accessing the network

isro2013 computer-networks network-security

Answer key☟

5.25 Network topologies (2)

5.25.1 Network topologies: ISRO CSE 2017 | Question: 27

Physical topology of FDDI is?

A. Bus
B. Ring
C. Star
D. None of the above

isro2017 computer-networks network-topologies

Answer key☟

5.25.2 Network topologies: ISRO CSE 2017 | Question: 30

If there are n devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively

A. $n(n-1)/2, n-1$
B. $n, n-1$
C. $n-1, n$
D. $n-1,\ n(n-1)/2$

isro2017 computer-networks network-topologies

Answer key☟

5.26 Out of Gatecse Syllabus (1)

5.26.1 Out of Gatecse Syllabus: GATE CSE 1993 | Question: 6.4, ISRO2008-14

Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through an asynchronous serial line at 2400 baud rate, and with two stop bits is

A. 109 B. 216 C. 218 D. 219

gate1993 computer-networks serial-communication normal isro2008 out-of-gatecse-syllabus

Answer key☟

5.27 Routers Bridge Hubs Switches (2)

5.27.1 Routers Bridge Hubs Switches: ISRO CSE 2011 | Question: 71

One SAN switch has 24 ports. All 24 supports 8 Gbps Fiber Channel technology. What is the aggregate bandwidth of that SAN switch?

A. 96 Gbps B. 192 Mbps C. 512 Gbps D. 192 Gbps

isro2011 computer-networks routers-bridge-hubs-switches

Answer key☟

5.27.2 Routers Bridge Hubs Switches: ISRO2015-48

A T-switch is used to

A. Control how messages are passed between computers
B. Echo every character that is received
C. Transmit characters one at a time
D. Rearrange the connections between computing equipments

isro2015 computer-networks routers-bridge-hubs-switches

Answer key☟

5.28 Routing (2)

5.28.1 Routing: ISRO CSE 2014 | Question: 16

What is routing algorithm used by OSPF routing protocol?

A. Distance vector B. Flooding C. Path vector D. Link state

computer-networks routing isro2014

Answer key☟

5.28.2 Routing: ISRO CSE 2016 | Question: 68

Dynamic routing protocol enable routers to

A. Dynamically discover and maintain routes
B. Distribute routing updates to other routers
C. Reach agreement with other routers about the network topology
D. All of the above

isro2016 computer-networks routing

Answer key☟

5.29 Sliding Window (3)

5.29.1 Sliding Window: GATE CSE 2009 | Question: 57, ISRO2016-75

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

A. $I = 2$ B. $I = 3$ C. $I = 4$ D. $I = 5$

gatecse-2009 computer-networks sliding-window normal isro2016

Answer key

5.29.2 Sliding Window: ISRO CSE 2008 | Question: 8

The TCP sliding window

A. can be used to control the flow of information
B. always occurs when the field value is 0
C. always occurs when the field value is 1
D. occurs horizontally

isro2008 computer-networks tcp sliding-window

Answer key

5.29.3 Sliding Window: ISRO CSE 2018 | Question: 44

Station$A$ uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is $80 \, ms$ and the bottleneck bandwidth on the path between A and B is $128 \, kbps$. What is the optimal window size that A should use?

a. 20 b. 40 c. 160 d. 320

isro2018 computer-networks sliding-window

Answer key

5.30 Slotted Aloha (1)

5.30.1 Slotted Aloha: ISRO2015-55

A certain population of ALOHA users manages to generate 70 request/sec. If the time is slotted in units of 50 msec, then channel load would be

A. 4.25 B. 3.5 C. 450 D. 350

isro2015 computer-networks slotted-aloha

Answer key

5.31 Stop and Wait (1)

5.31.1 Stop and Wait: ISRO-2013-41

What will be the efficiency of a Stop and Wait protocol, if the transmission time for a frame is 20ns and the propagation time is 30ns?

A. 20% B. 25% C. 40% D. 66%

isro2013 computer-networks stop-and-wait

Answer key

5.32 Subnetting (11)

5.32.1 Subnetting: GATE CSE 2003 | Question: 82, ISRO2009-1

The subnet mask for a particular network is $255.255.31.0$. Which of the following pairs of IP addresses could belong to this network?

A. $172.57.88.62$ and $172.56.87.23$
B. $10.35.28.2$ and $10.35.29.4$
C. $191.203.31.87$ and $191.234.31.88$
D. $128.8.129.43$ and $128.8.161.55$

gatecse-2003 computer-networks subnetting normal isro2009

Answer key☟

5.32.2 Subnetting: GATE CSE 2007 | Question: 67, ISRO2016-72

The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

A. 62 subnets and 262142 hosts.
B. 64 subnets and 262142 hosts.
C. 62 subnets and 1022 hosts.
D. 64 subnets and 1024 hosts.

gatecse-2007 computer-networks subnetting easy isro2016

Answer key☟

5.32.3 Subnetting: GATE CSE 2012 | Question: 34, ISRO-DEC2017-32

An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: $245.248.128.0/20$. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

A. $245.248.136.0/21 \text{ and } 245.248.128.0/22$
B. $245.248.128.0/21 \text{ and } 245.248.128.0/22$
C. $245.248.132.0/22 \text{ and } 245.248.132.0/21$
D. $245.248.136.0/24 \text{ and } 245.248.132.0/21$

gatecse-2012 computer-networks subnetting normal isrodec2017

Answer key☟

5.32.4 Subnetting: GATE IT 2006 | Question: 63, ISRO2015-57

A router uses the following routing table:

Destination	**Mask**	**Interface**
144.16.0.0	255.255.0.0	eth0
144.16.64.0	255.255.224.0	eth1
144.16.68.0	255.255.255.0	eth2
144.16.68.64	255.255.255.224	eth3

Packet bearing a destination address $144.16.68.117$ arrives at the router. On which interface will it be forwarded?

A. eth0
B. eth1
C. eth2
D. eth3

gateit-2006 computer-networks subnetting normal isro2015

Answer key☟

5.32.5 Subnetting: ISRO CSE 2007 | Question: 73

Range of IP Address from $224.0.0.0$ to $239.255.255.255$ are

A. Reserved for loopback
B. Reserved for broadcast
C. Used for multicast packets
D. Reserved for future addressing

isro2007 computer-networks subnetting

Answer key☟

5.32.6 Subnetting: ISRO CSE 2008 | Question: 3

The subnet mask $255.255.255.192$

A. extends the network portion to 16 bits
B. extends the network portion to 26 bits
C. extends the network portion to 36 bits
D. has no effect on the network portion of an IP address

isro2008 computer-networks subnetting

Answer key☟

5.32.7 Subnetting: ISRO CSE 2011 | Question: 45

The broadcast address for IP network $172.16.0.0$ with subnet mask $255.255.0.0$ is

A. $172.16.0.255$
B. $172.16.255.255$
C. $255.255.255.255$
D. $172.255.255.255$

isro2011 computer-networks subnetting

Answer key☟

5.32.8 Subnetting: ISRO CSE 2014 | Question: 75

An organization is granted the block $130.34.12.64/26$. It needs to have 4 subnets. Which of the following is not an address of this organization?

A. $130.34.12.124$
B. $130.34.12.89$
C. $130.34.12.70$
D. $130.34.12.132$

computer-networks subnetting isro2014

Answer key☟

5.32.9 Subnetting: ISRO CSE 2017 | Question: 29

The default subnet mask for a class B network can be

A. $255.255.255.0$
B. $255.0.0.0$
C. $255.255.192.0$
D. $255.255.0.0$

isro2017 computer-networks subnetting

Answer key☟

5.32.10 Subnetting: ISRO-2013-43

What is IP class and number of sub-networks if the subnet mask is $255.224.0.0$?

A. Class A, 3
B. Class A, 8
C. Class B, 3
D. Class B, 32

isro2013 computer-networks subnetting

Answer key

5.32.11 Subnetting: ISRO2015-54

In a class B subnet, we know the IP address of one host and the mask as given below:

IP address : $125.134.112.66$

Mask : $255.255.224.0$

What is the first address(Network address)?

A. $125.134.96.0$
B. $125.134.112.0$
C. $125.134.112.66$
D. $125.134.0.0$

isro2015 computer-networks subnetting

Answer key

5.33 Supernetting (1)

5.33.1 Supernetting: ISRO CSE 2014 | Question: 57

A supernet has a first address of $205.16.32.0$ and a supernet mask of $255.255.248.0$. A router receives 4 packets with the following destination addresses. Which packet belongs to this supernet?

A. $205.16.42.56$
B. $205.17.32.76$
C. $205.16.31.10$
D. $205.16.39.44$

isro2014 computer-networks supernetting

Answer key

5.34 TCP (6)

5.34.1 TCP: ISRO CSE 2007 | Question: 66

Silly Window Syndrome is related to

A. Error during transmission
B. File transfer protocol
C. Degrade in TCP performance
D. Interface problem

isro2007 computer-networks tcp

Answer key

5.34.2 TCP: ISRO CSE 2014 | Question: 60

Suppose you are browsing the world wide web using a web browser and trying to access the web servers. What is the underlying protocol and port number that are being used?

A. UDP, 80
B. TCP, 80
C. TCP, 25
D. UDP, 25

isro2014 computer-networks tcp

Answer key

5.34.3 TCP: ISRO CSE 2020 | Question: 53

The persist timer is used in TCP to

A. To detect crashes from the other end of the connection
B. To enable retransmission
C. To avoid deadlock condition
D. To timeout FIN_Wait1 condition

isro-2020 computer-networks tcp normal

Answer key

5.34.4 TCP: ISRO CSE 2020 | Question: 54

Checksum field in TCP header is

A. ones complement of sum of header and data in bytes
B. ones complement of sum of header, data and pseudo header in 16 bit words
C. dropped from IPv6 header format
D. better than md5 or sh1 methods

isro-2020 computer-networks tcp normal

Answer key

5.34.5 TCP: ISRO2015-50

How many bits internet address is assigned to each host on a TCP/IP internet which is used in all communication with the host?

A. 16 bits B. 32 bits C. 48 bits D. 64 bits

isro2015 computer-networks tcp

Answer key

5.34.6 TCP: ISRO2015-53

An ACK number of 1000 in TCP always means that

A. 999 bytes have been successfully received
B. 1000 bytes have been successfully received
C. 1001 bytes have been successfully received
D. None of the above

isro2015 computer-networks tcp

Answer key

5.35 Throughput (2)

5.35.1 Throughput: ISRO CSE 2023 | Question: 44

A host in one org other organisation using another host with 25 m sec RTT. Network 1 Gbps link of this network tuned the administrator 1000 bytes to transfer large packet size to files optimally ovel utilisation $> 96\%$, how big to have chann (in terms of packets) should window size (assuming ACK packets have no be (assuming ACK pals impact)?

A. 2000 packets
B. 2400 packets
C. 3000 packets
D. 4000 packets

isro-cse-2023 computer-networks sliding-window throughput

5.35.2 Throughput: ISRO-DEC2017-36

Station A uses $32-byte$ packets to transmit messages to Station B using a sliding window protocol.
The round trip time delay between A and B is $40\,ms$ and the bottleneck bandwidth on the path A and B is $64\,kbps$.What is the optimal window size that A should use?

A. 5 B. 10 C. 40 D. 80

isrodec2017 computer-networks sliding-window throughput

Answer key

5.36 Token Ring (1)

5.36.1 Token Ring: GATE CSE 2007 | Question: 66, ISRO2016-71

In a token ring network the transmission speed is 10^7 bps and the propagation speed is $200 \text{ meters}/\mu s$. The 1-bit delay in this network is equivalent to:

A. 500 meters of cable.
B. 200 meters of cable.
C. 20 meters of cable.
D. 50 meters of cable.

gatecse-2007 computer-networks token-ring out-of-syllabus-now isro2016

Answer key☟

5.37 Transport Layer (1)

5.37.1 Transport Layer: ISRO CSE 2018 | Question: 45

Assuming that for a given network layer implementation, connection establishment overhead is 100 bytes and disconnection overhead is 28 bytes. What would be the minimum size of the packet the transport layer needs to keep up, if it wishes to implement a datagram service above the network layer and needs to keep its overhead to a minimum of 12.5%. (ignore transport layer overhead)

a. 512 bytes
b. 768 bytes
c. 1152 bytes
d. 1024 bytes

isro2018 transport-layer

Answer key☟

5.38 UDP (1)

5.38.1 UDP: ISRO-DEC2017-34

Generally, TCP is reliable and UDP is not reliable. DNS which has to be reliable uses UDP because

A. UDP is slower.
B. DNS servers has to keep connections.
C. DNS requests are generally very small and fit well within UDP segments.
D. None of these.

isrodec2017 computer-networks tcp udp

Answer key☟

5.39 Wifi (1)

5.39.1 Wifi: ISRO CSE 2017 | Question: 38

What is WPA?

A. wired protected access
B. wi-fi protected access
C. wired process access
D. wi-fi process access

isro2017 computer-networks wifi

Answer key☟

5.40 Wireless Lan (1)

5.40.1 Wireless Lan: ISRO CSE 2023 | Question: 66

Match the following :

	Versions of 802.11		Speed of Wireless LAN
(i)	$802.11ac$	(P)	7 Gbps
(ii)	$802.11a/g$	(Q)	11 Mbps
(iii)	$802.11ad$	(R)	54 Gbps
(iv)	$802.11b$	(S)	3.5 Mbps

A. (i) - (S), (ii) - (R), (iii) - (P), (iv) - (Q)
B. (i) - (S), (ii) −(R), (iii) −(Q), (iv) - (P)
C. (i) - (R), (ii) - (S), (iii) - (P), (iv) - (Q)
D. (i) −(R), (ii) −(Q), (iii) −(P), (iv) - (S)

isro-cse-2023 computer-networks wireless-lan network-protocols

Answer key☟

5.41 Wireless Networks (1)

5.41.1 Wireless Networks: ISRO CSE 2007 | Question: 74

IEEE 802.11 is standard for

A. Ethernet
B. Bluetooth
C. Broadband Wireless
D. Wireless LANs

isro2007 computer-networks wireless-networks

Answer key☟

Answer Keys

5.0.1	A	5.0.2	C	5.0.3	D	5.0.4	D	5.0.5	C
5.0.6	A	5.0.7	D	5.0.8	N/A	5.0.9	C	5.0.10	A
5.0.11	N/A	5.1.1	A	5.2.1	B	5.2.2	B	5.2.3	C
5.2.4	B	5.3.1	A	5.3.2	C	5.4.1	D	5.4.2	C
5.4.3	A	5.4.4	C	5.4.5	D	5.4.6	D	5.4.7	D
5.5.1	B	5.5.2	B	5.5.3	C	5.5.4	A	5.5.5	B
5.5.6	A	5.5.7	C	5.5.8	C	5.6.1	D	5.7.1	A
5.8.1	C	5.8.2	C	5.9.1	D	5.9.2	B	5.10.1	C
5.10.2	C	5.11.1	A	5.11.2	A	5.11.3	C	5.11.4	6

5.12.1	A
5.12.6	A
5.15.1	B
5.18.1	C
5.20.4	B
5.22.3	A
5.23.5	B
5.23.10	C
5.24.4	C
5.24.9	C
5.25.2	A
5.28.2	D
5.31.1	B
5.32.5	C
5.32.10	B
5.34.3	C
5.35.2	B
5.40.1	A

5.12.2	B
5.13.1	A
5.15.2	C
5.19.1	B
5.21.1	B
5.23.1	D
5.23.6	A
5.23.11	C
5.24.5	D
5.24.10	B
5.26.1	C
5.29.1	D
5.32.1	D
5.32.6	B
5.32.11	A
5.34.4	B;C
5.36.1	C
5.41.1	D

5.12.3	D
5.14.1	C
5.15.3	D
5.20.1	D
5.21.2	C
5.23.2	C
5.23.7	D
5.24.1	D
5.24.6	D
5.24.11	A
5.27.1	B
5.29.2	A
5.32.2	C
5.32.7	B
5.33.1	D
5.34.5	B
5.37.1	D

5.12.4	D
5.14.2	B
5.16.1	C
5.20.2	A
5.22.1	B
5.23.3	D
5.23.8	A
5.24.2	D
5.24.7	A
5.24.12	D
5.27.2	D
5.29.3	B
5.32.3	A
5.32.8	D
5.34.1	C
5.34.6	D
5.38.1	C

5.12.5	A
5.14.3	D
5.17.1	C
5.20.3	D
5.22.2	D
5.23.4	B
5.23.9	A
5.24.3	B
5.24.8	A
5.25.1	B
5.28.1	D
5.30.1	B
5.32.4	C
5.32.9	D
5.34.2	B
5.35.1	C
5.39.1	B